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EDUCATION

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**NATIONAL
SENIOR CERTIFICATE/
NASIONALE
SENIORSERTIFIKAAT**

GRADE/GRAAD 11

NOVEMBER 2024

**PHYSICAL SCIENCES P2
MARKING GUIDELINE/
FISIESE WETENSKAPPE V2
NASIENRIGLYN**

MARKS/PUNTE: 150

This marking guideline consists of 11 pages./
Hierdie nasienriglyn bestaan uit 11 bladsye.

QUESTION/VRAAG 1

- | | | |
|------|------|-------------|
| 1.1 | D ✓✓ | (2) |
| 1.2 | C ✓✓ | (2) |
| 1.3 | C ✓✓ | (2) |
| 1.4 | B ✓✓ | (2) |
| 1.5 | B ✓✓ | (2) |
| 1.6 | B ✓✓ | (2) |
| 1.7 | D ✓✓ | (2) |
| 1.8 | B ✓✓ | (2) |
| 1.9 | A ✓✓ | (2) |
| 1.10 | D ✓✓ | (2) |
| | | [20] |

QUESTION/VRAAG 2

- 2.1 The sharing of electrons between two atoms to form a molecule. ✓✓
 Die deel van elektrone tussen atome om 'n molekule te vorm. ✓✓ (2)
- 2.2. 2.2.1 I_2 ✓ (1)
- 2.2.2 KBr ✓ (1)
- 2.2.3 NH_3 ✓ (1)
- 2.2.4 I_2 ✓ (1)
- 2.3 2.3.1 $H:\ddot{O}:Cl:$ ✓✓ (2)
- 2.3.2 $H:S:H$ ✓✓ (2)
- 2.4 2.4.1 Dative covalent bond / coordinate covalent bond ✓
 Datiewe kovalente binding / koördinate-kovalente binding ✓ (1)
- 2.4.2
$$H:\ddot{N}:H \checkmark + H^+ \checkmark \rightarrow \left[\begin{array}{c} H \\ H:\ddot{N}:H \\ H \end{array} \right]^+ \checkmark \checkmark$$
 (4)
- 2.5 2.5.1 Bent / Gebuig ✓ (1)
- 2.5.2 Trigonal pyramidal / Trigonaal piramidaal ✓ (1)
- 2.6 Yes / Ja ✓ (1)
- 2.7 I_2 is a non-polar molecule and only contains London forces. ✓
 CCl_4 is a non-polar molecule and only contains London forces. ✓
 Both compounds are non-polar and only has London forces and "like dissolves like" ✓

 I_2 is 'n niet-polêre molekule en bevat slegs London-kragte ✓
 CCl_4 is 'n niet-polêre molekule en bevat slegs London-kragte ✓
 Beide verbindings is niet-polêr en bevat slegs London-kragte en "soort los soort op" ✓ (3)
- 2.8 2.8.1 Bond energy is the energy needed to break one mole of its molecules into separate atoms. ✓✓
 Bindingsenergie is die energie benodig om een mol van sy molekule in aparte atome op te breek. ✓✓ (2)

- 2.8.2 The atomic size decreases in period (from C to F).
 The bond length decreases the C to F atom. ✓
 The shorter the bond length the more energy is needed to break them ✓
Die atomiesegrootte neem af in 'n periode (vanaf C na F).
Die bindingslengte neem vanaf C na F atoom af. ✓
Hoe korter die bindinglengte hoe meer energie word benodig om dit te breek ✓

(2)
[25]

QUESTION/VRAAG 3

- 3.1 The temperature at which the vapour pressure of a substance equals atmospheric pressure. ✓✓
Die temperatuur waarby die dampdruk van 'n stof gelyk aan die atmosferiese druk is. ✓✓ (2)
- 3.2 YES./ JA ✓
 Only one independent variable / All variables are controlled except molecular mass ✓
Slegs een onafhanklike veranderlike / Alle veranderlike behalwe vir molekulêremassa is beheer ✓ (2)
- 3.3 Gas ✓ (1)
- 3.4
- The molecular mass increases from A to D ✓
 - The strength of the London forces increases with an increase in the molecular size/mass. ✓
 - More energy will be required to overcome the intermolecular forces from A to D ✓
 - *Die molekulêre massa neem toe vanaf A to D*
 - *Die sterkte van die London-kragte neem toe met 'n toename in die molekulêremassa/grootte* ✓
 - *Meer energie word benodig om die intermolekulêrekragte vanaf A tot D te oorkom* ✓

OR/OF

- The molecular mass decreases from D to A ✓
 - The strength of the London forces decreases with decrease in molecular size/mass. ✓
 - Less energy will be required to overcome the intermolecular forces from D to A ✓
 - *Die molekulêre massa neem af vanaf D tot A* ✓
 - *Die sterkte van die London-kragte neem af soos die molekulêremassa / grootte afneem* ✓
 - *Minder energie word benodig om die intermolekulêrekragte vanaf D tot A te oorkom* ✓
- (3)

3.5 Compound A / *Verbinding A* CH_4 ✓

It has the lowest boiling point / *Dit het die laagste kookpunt* ✓ (2)

3.6 3.6.1 Lower than / *Laer as* ✓

(1)

3.6.2 • HF has hydrogen bonds ✓

• HCl has dipole dipole forces ✓

• Hydrogen bonds are stronger than dipole dipole forces ✓

• More energy is needed to overcome intermolecular forces in HF ✓

• HF het waterstofbinding

• HCl het dipool-dipoolkragte

• Waterstofbinding is sterker as die dipool-dipoolkragte

• Meer energie word benodig om die intermolekulêre kragte in HF te oorkom

OR / OF

• HF has hydrogen bonds ✓

• HCl has dipole dipole forces ✓

• Dipole dipole forces is weaker than Hydrogen bonds ✓

• Less energy is needed to overcome intermolecular forces in HCl ✓

• HF het waterstofbinding ✓

• HCl het dipool-dipoolkragte ✓

• Dipool-dipoolkragte is swakker as die waterstofbinding ✓

Minder energie word benodig om die intermolekulêre kragte in HCl te oorkom ✓

(4)

[15]

QUESTION/VRAAG 4

4.1 Boyle's law ✓

The pressure of an enclosed gas is inversely proportional to the volume it occupies at constant temperature. ✓✓

Boyle se wet ✓

Die druk van 'n ingeslote gas is omgekeerd eweredig aan die volume wat dit by konstante temperatuur beslaan. ✓✓

(3)

4.2 **Criteria for hypothesis / Nasienkriteria vir hipotese**

The independent and dependent variables are stated correctly ✓

Die onafhanklike en afhanklike veranderlike korrek gestel ✓

A Statement about the relationship between the independent and dependent variables ✓

'n Stelling rondom die verwantskap tussen onafhanklike en afhanklike veranderlike ✓

The higher the pressure of a gas the lower its volume ✓✓

Hoe hoër die druk van 'n gas hoe laer is sy volume ✓✓

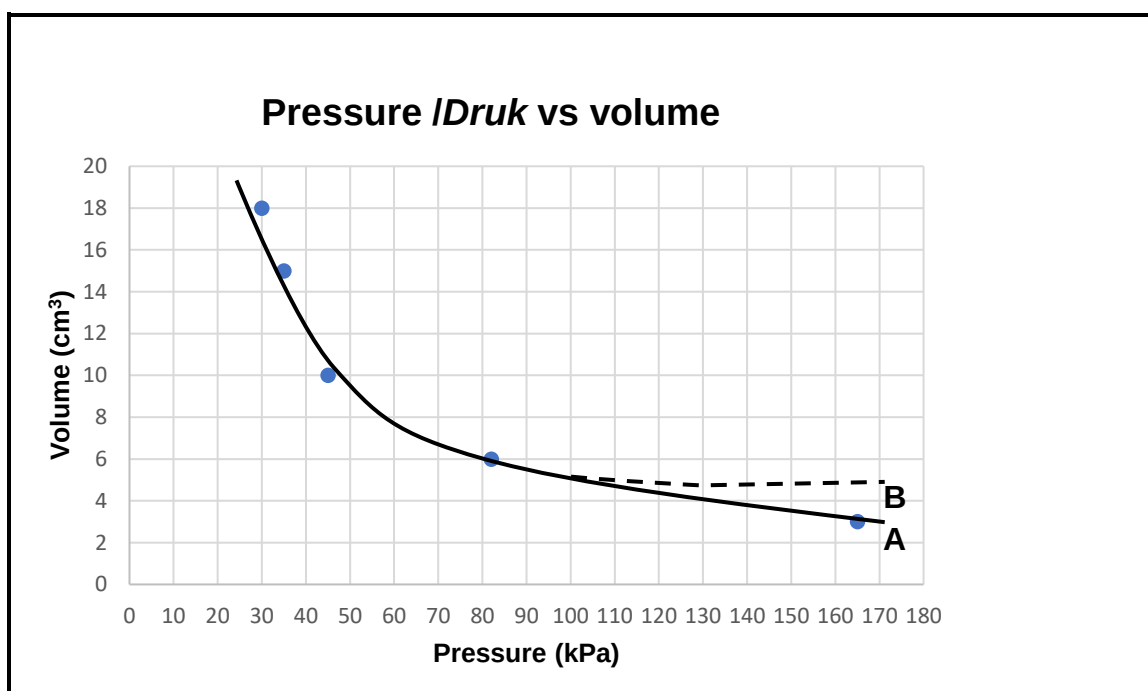
(2)

4.3 $40 \times 10^3 \text{ Pa}$ ✓ /4 kPa (1)

4.4 $p_1V_1 = p_2V_2$ ✓
 $(40 \times 10^3) (12) \checkmark = (200 \times 10^3) V_2 \checkmark$
 $V_2 = 2,4 \text{ cm}^3 \checkmark$
 (Use any correct co-ordinates from the graph / Gebruik enige korrekte koördinate vanaf die grafiek) (4)

4.5 **Marking Criteria/ Nasienkriteria**

Curve of **B** is higher than curve **A** at high pressure ✓✓
 Kurwe van **B** is hoër as die kurwe van **A** by hoë druk ✓✓



(2)
[12]

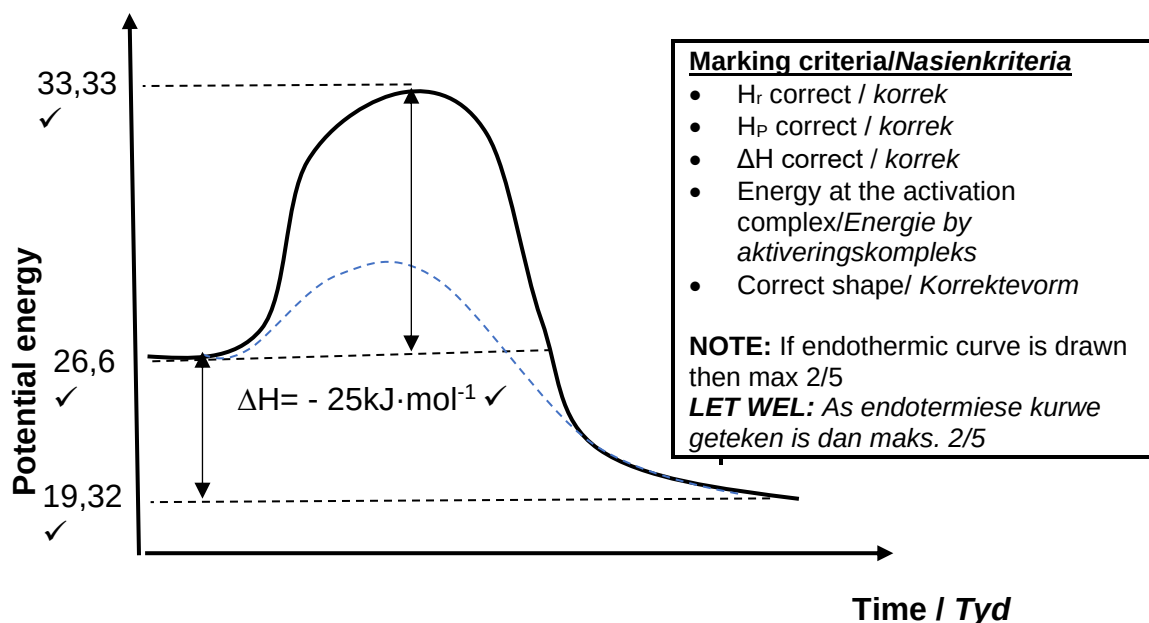
QUESTION/VRAAG 5

5.1 The minimum energy needed to for a reaction to take place. ✓✓
 Die minimum energie wat benodig word vir 'n reaksie om plaas te vind. ✓✓ (2)

5.2 Exothermic / Eksotermies. ✓ $\Delta H < 0$ ✓ (2)

5.3 $\Delta H = H_p - H_r$
 $-7,28 = H_p - 26,6 \checkmark$
 $H_p = 19,32 \text{ kJ} \cdot \text{mol}^{-1} \checkmark$ (2)

5.4



(5)

5.5 See sketch Question 5.4 / Sien skets VRAAG 5.4

Marking criteria / Nasienkriteria

- Curve start at the initial potential energy of reactants and end at the final potential energy of the products. / Kurwe begin die aanvangs potensiële energie van die reaktanse en eindig by die finale potensiële energie van die produkte ✓
- Curve peak is lower / Kurwe piek is laer ✓

(2)

[13]**QUESTION/VRAAG 6**

6.1 6.1.1 The simplest whole number ratio of atoms in a compound. ✓✓

Die eenvoudigste heelgetal verhouding van atome in verbinding. ✓✓

(2)

$$6.1.2 \quad \underline{P(58) + 13(32)} \checkmark = \underline{8(44) + 10(18)} \checkmark$$

$$P = 2 \checkmark$$

(3)

6.1.3 **OPTION 1: Using percentage composition/ OPSIE 1: Gebruik die persentasie samestelling**

$$\begin{array}{lcl} \text{Mol C} & : & \text{Mol H} \\ \frac{82,76}{12} & \checkmark : & \frac{17,24}{1} \checkmark \end{array}$$

$$6,90 : 17,24$$

$$1 : 2,5$$

$$2 : 5 \checkmark$$

Marking criteria/Nasienkriteria

- % C divide by M (C)
% C gedeel deur M (C)
- % H divide by M (H)
% H gedeel deur M(H)
- Simplest mole ratio
Eenvoudigste molverhouding
- Ratio / verhouding
- Molecular formula
Molekulêre formule

Empirical formula / Empiriese formule : C_2H_5 Formula mass / Formule massa = $2(12) + 1(5) = 29$ Ratio / Verhouding = $58 / 29 = 2 \checkmark$

Molecular formula / Molekulêre formule = C_4H_{10} ✓**OPTION 2: Using conservation of mass/ OPSIE 2 : Gebruik die behoud van massa**

Number of carbon reactants / Aantal koolstowwe in reaktanse =

Number of carbon products / Aantal koolstowwe in produkte

$$2 C = 8 C \quad \checkmark$$

$$C = 4$$

Number of hydrogens reactants/ Aantal waterstofatome in

reaktanse = Number of hydrogens products / Aantal

waterstofatome in produkte

$$2 H = 10 H_2 \quad \checkmark$$

$$H = (10)(2) / 2 \quad \checkmark$$

$$H = 10$$

✓ Both / Beide

Molecular formula / Molekulêre formule = C_4H_{10} ✓

(5)

6.2

6.2.1

Limiting reagent is a reactant that is completely used up in a chemical reaction. ✓✓

Die beperkende reagens is die stof wat volledig opgebruik word tydens 'n chemiese reaksie. ✓✓

(2)

6.2.2

$$n = \frac{m}{M} \quad \checkmark$$

$$n = \frac{5}{106} \quad \checkmark$$

$$n = 0,047 \text{ mol}$$

$$n(HCl) = 2(0,047) \quad \checkmark$$

$$n(HCl) = 0,094 \text{ mol} \quad \checkmark$$

(4)

6.2.3

$$n(Na_2CO_3) = n(CO_2) = 0,047 \text{ mol} \quad \checkmark$$

$$V = nV_m \quad \checkmark$$

$$V = (0,047)(24,45) \quad \checkmark$$

$$V = 1,149 \text{ dm}^3 \quad \checkmark$$

(4)

6.2.4 $n(\text{HCl}) \text{ initial / aanvangs} = 0,094 \times 100 / 76 \checkmark = 0,124 \text{ mol}$

$$c = \frac{n}{V} \checkmark$$

$$c = \frac{0,124}{0,1} \checkmark$$

$$c = 1,24 \text{ mol} \cdot \text{dm}^{-3} \checkmark$$

$$c = \frac{n}{V} \checkmark$$

$$c = \frac{0,124}{0,25}$$

$$c = 0,496 \text{ mol} \cdot \text{dm}^{-3}$$

$$c_1 V_1 = c_2 V_2$$

$$(0,496)(250) = c_2(100) \checkmark$$

$$c_2 = 1,24 \text{ mol} \cdot \text{dm}^{-3} \checkmark$$

(4)

[24]

QUESTION/VRAAG 7

7.1 7.1.1 Au Acid is a substance that produces hydrogen ions (H^+)/ hydronium ions (H_3O^+) when it dissolves in water. $\checkmark\checkmark$
'n Suur is 'n stof wat waterstofione (H^+)/ hidroniumione (H_3O^+) vorm wanneer dit in water oplos. $\checkmark\checkmark$ (2)

7.1.2 $\text{H}_2\text{O} \checkmark$ and/ en $\text{HCO}_3^- \checkmark$ (2)

7.1.3 $\text{H}_2\text{CO}_3 \checkmark$ and/ en $\text{H}_3\text{O}^+ \checkmark$ (2)

7.1.4 $\text{H}_2\text{CO}_3 + 2 \text{NaOH} \checkmark \rightarrow \text{Na}_2\text{CO}_3 + 2 \text{H}_2\text{O} \checkmark$ ($\checkmark$ bal) (3)

7.2 7.2.1

OPTION 1/ OPSIE 1

$$c = \frac{m}{MV} \checkmark$$

$$c = \frac{2}{(40)(0,1)} \checkmark$$

$$c = 0,5 \text{ mol} \cdot \text{dm}^{-3} \checkmark$$

OPTION 2 / OPSIE 2

$$n = \frac{m}{M}$$

$$n = \frac{2}{40}$$

$$n = 0,05$$

$$c = \frac{n}{V} \checkmark$$

$$c = \frac{0,05}{0,1} \checkmark$$

$$c = 0,5 \text{ mol} \cdot \text{dm}^{-3} \checkmark$$

(3)

7.2.2

OPTION 1 / OPSIE 1

$$[\text{NaOH}] = [\text{OH}^-]$$

$$[\text{OH}^-] = 0,5 \text{ mol} \cdot \text{dm}^{-3}$$

$$[\text{OH}^-][\text{H}_3\text{O}^+] = 1 \times 10^{-14}$$

$$[\text{H}_3\text{O}^+] (0,5) = 1 \times 10^{-14} \checkmark$$

$$[\text{H}_3\text{O}^+] = 2 \times 10^{-14} \text{ mol} \cdot \text{dm}^{-3}$$

$$\text{pH} = -\log[\text{H}_3\text{O}^+] \checkmark$$

$$= -\log(2 \times 10^{-14}) \checkmark$$

$$= 13,70 \checkmark$$

OPTION 1 / OPSIE 1

$$\text{pOH} = -\log[\text{OH}^-]$$

$$\text{pOH} = -\log(0,5) \checkmark$$

$$\text{pOH} = 0,30$$

$$\text{pH} + \text{pOH} = 14 \checkmark$$

$$\text{pH} + 0,30 = 14 \checkmark$$

$$\text{pH} = 13,70 \checkmark$$

(4)

7.3 7.3.1

$$n = \frac{N}{N_A} \checkmark$$

$$= \frac{9,033 \times 10^{22}}{6,02 \times 10^{23}} \checkmark$$

$$n = 0,15 \text{ mol} \checkmark$$

(3)

7.3.2

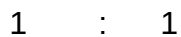
Positive marking from QUESTION 7.3.1/ Positiewe nasien vanaf VRAAG 7.3.1**Marking criteria / Nasienkriteria**

- Formula / Formule $c = n/V$
- Subst. of NaOH values into / Vervang waardes van NaOH in $n = cV$
- **Using** mol ratio/ **Gebruik** die molverhouding $\text{HCl} : \text{NaOH} = 1 : 1$
- Determine the mol of HCl in reaction 1 / Bepaal die mol HCl in reaksie 1
- **Using** ratio / **Gebruik** die molverhouding $\text{HCl} : \text{MgCO}_3 = 2 : 1$
- Subst. into / Vervang in $m = nM$
- Subst. into percentage formula / Vervang in persentasie formule
- Final answer / Finale antwoord

$$n(\text{NaOH}) = cV \checkmark$$

$$n(\text{NaOH}) = 0,8 \times \frac{55}{1000} \checkmark$$

$$n = 0,044 \text{ mol}$$

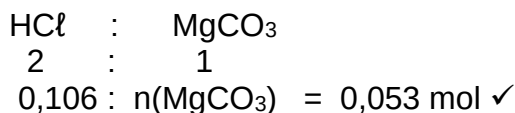


$$n(\text{HCl})_{\text{excess}} = 0,044 \text{ mol} \checkmark$$

$$n(\text{HCl})_{\text{reaction 1}} = n(\text{HCl})_{\text{initial}} - n(\text{HCl})_{\text{excess}}$$

$$= 0,15 - 0,044 \checkmark$$

$$= 0,106 \text{ mol}$$



$$\begin{aligned} m &= nM \\ &= 0,053 \times 84 \checkmark \\ &= 4,452 \text{ g} \end{aligned}$$

$$\begin{aligned} \% \text{ purity} &= \frac{\text{pure mass}}{\text{Sample mass}} \times 100\% \\ &= \frac{4,452}{5} \times 100 \% \checkmark \\ &= 89,04 \% \checkmark \end{aligned}$$

(8)
[27]**QUESTION/VRAAG 8**

- 8.1 8.1.1 A redox is a reaction that involves an electron transfer. ✓✓
'n Redoksreaksie is 'n reaksie wat 'n elektronoordrag behels. ✓✓

OR / OF

It is a reaction that always involve changes in oxidation numbers. ✓✓
Dit is 'n reaskie wat altyd 'n verandering in oksidasiegetalle behels. ✓✓ (2)

- 8.1.2 Ni^{2+} ✓

Oxidation number decreases from +2 to 0. ✓✓
Die oksidasiegetal neem af vanaf +2 na 0. ✓✓ (3)

- 8.1.3 Nitrate ion (NO_3^-) is spectator ion / undergoes no changes in its oxidation state. ✓
Nitraatioon (NO_3^-) is 'n toeskouerioon/ ondergaan geen verandering in sy oksidasiegetal (1)

- 8.2 8.2.1 $\text{Cr}_2\text{O}_7^{2-}$
 $2x + 7(-2) = -2$ ✓
 $x = +6$ ✓ (2)

- 8.2.2 $\text{Cr}_2\text{O}_7^{2-} + 14 \text{H}^+ + 6\text{e}^- \rightarrow 2 \text{Cr}^{3+} + 7 \text{H}_2\text{O}$ ✓✓ (2)

- 8.2.3 $\text{Sn}^{2+} \rightarrow \text{Sn}^{4+} + 2\text{e}^-$ ✓
 $\text{Cr}_2\text{O}_7^{2-} + 14 \text{H}^+ + 6\text{e}^- \rightarrow 2 \text{Cr}^{3+} + 7 \text{H}_2\text{O}$
 $3 \text{Sn}^{2+} + \text{Cr}_2\text{O}_7^{2-} + 14 \text{H}^+ \checkmark \rightarrow 3 \text{Sn}^{4+} + 2 \text{Cr}^{3+} + 7 \text{H}_2\text{O} \checkmark$ (✓ bal) (4)
[14]

TOTAL/TOTAAL: 150